

Given: Due:	Further Maths (Statistics) HW 13	Name:												
Chapter 1: Discrete Random Variables														
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.														
91% (A*)	76% (A)	66% (B)												
56% (C)	50% (D)	40% (E)												
<i>Useful resources:</i> Your notes from recent lessons & your text book														
Questions - do these on lined paper														
1) A fair blue die has faces numbered 1, 1, 3, 3, 5 and 5. The random variable B represents the score when the blue die is rolled.														
(a) Write down the probability distribution for B . (2)														
(b) State the name of this probability distribution. (1)														
(c) Write down the value of $E(B)$. (1)														
A second die is red and the random variable R represents the score when the red die is rolled. The probability distribution of R is														
<table border="1" style="margin: auto; border-collapse: collapse;"> <tr> <td style="padding: 5px;">r</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">4</td> <td style="padding: 5px;">6</td> </tr> <tr> <td style="padding: 5px;">$P(R = r)$</td> <td style="padding: 5px;">$\frac{2}{3}$</td> <td style="padding: 5px;">$\frac{1}{6}$</td> <td style="padding: 5px;">$\frac{1}{6}$</td> </tr> </table>			r	2	4	6	$P(R = r)$	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$				
r	2	4	6											
$P(R = r)$	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$											
(d) Find $E(R)$. (2)														
(e) Find $\text{Var}(R)$. (3)														
Tom invites Avisha to play a game with these dice. Tom spins a fair coin with one side labelled 2 and the other side labelled 5. When Avisha sees the number showing on the coin she then chooses one of the dice and rolls it. If the number showing on the die is <u>greater</u> than the number showing on the coin, Avisha wins, otherwise Tom wins. Avisha chooses the die which gives her the best chance of winning each time Tom spins the coin.														
(f) Find the probability that Avisha wins the game, stating clearly which die she should use in each case. (4)														
2) The score S when a spinner is spun has the following probability distribution.														
<table border="1" style="margin: auto; border-collapse: collapse;"> <tr> <td style="padding: 5px;">s</td> <td style="padding: 5px;">0</td> <td style="padding: 5px;">1</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">4</td> <td style="padding: 5px;">5</td> </tr> <tr> <td style="padding: 5px;">$P(S = s)$</td> <td style="padding: 5px;">0.2</td> <td style="padding: 5px;">0.2</td> <td style="padding: 5px;">0.1</td> <td style="padding: 5px;">0.3</td> <td style="padding: 5px;">0.2</td> </tr> </table>			s	0	1	2	4	5	$P(S = s)$	0.2	0.2	0.1	0.3	0.2
s	0	1	2	4	5									
$P(S = s)$	0.2	0.2	0.1	0.3	0.2									
(a) Find $E(S)$. (2)														
(b) Show that $E(S^2) = 10.4$ (2)														
(c) Hence find $\text{Var}(S)$. (2)														
(d) Find (i) $E(5S - 3)$, (ii) $\text{Var}(5S - 3)$. (4)														
(e) Find $P(5S - 3 > S + 3)$ (3)														
The spinner is spun twice.														

The score from the first spin is S_1 and the score from the second spin is S_2 The random variables S_1 and S_2 are independent and the random variable $X = S_1 \times S_2$ (f) Show that $P(S_1 = 1 \cap X < 5) = 0.16$ (g) Find $P(X < 5)$.		(2)												
3) The random variable X has probability distribution <table border="1"> <tr> <td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td></tr> <tr> <td>$P(X = x)$</td><td>0.2</td><td>p</td><td>0.2</td><td>q</td><td>0.15</td></tr> </table> (a) Given that $E(X) = 4.5$, write down two equations involving p and q. Find (b) the value of p and the value of q, (c) $P(4 < X \leq 7)$. Given that $E(X^2) = 27.4$, find (d) $\text{Var}(X)$, (e) $E(19 - 4X)$, (f) $\text{Var}(19 - 4X)$.	x	1	3	5	7	9	$P(X = x)$	0.2	p	0.2	q	0.15		(3)
x	1	3	5	7	9									
$P(X = x)$	0.2	p	0.2	q	0.15									
4) A fair six-sided black die has faces numbered 1, 2, 2, 3, 3 and 4 The random variable B represents the score when the black die is rolled. (a) Write down the value of $E(B)$ A white die has 6 faces numbered 1, 1, 2, 4, 5 and c where $c > 5$ The discrete random variable W represents the score when the white die is rolled and has probability distribution given by <table border="1"> <tr> <td>w</td><td>1</td><td>2</td><td>4</td><td>5</td><td>c</td></tr> <tr> <td>$P(W = w)$</td><td>$a + b$</td><td>a</td><td>0.3</td><td>a</td><td>b</td></tr> </table> Greg and Nilaya play a game with these dice. Greg throws the black die and Nilaya throws the white die. Greg wins the game if he scores at least two more than Nilaya, otherwise Greg loses. The probability of Greg winning the game is $\frac{1}{6}$ (b) Find the value of a and the value of b Show your working clearly. The random variable $X = 2W - 5$ Given that $E(X) = 2.6$ (c) find the exact value of $\text{Var}(X)$	w	1	2	4	5	c	$P(W = w)$	$a + b$	a	0.3	a	b		(1)
w	1	2	4	5	c									
$P(W = w)$	$a + b$	a	0.3	a	b									
		(5)												
		(6)												

END OF HOMEWORK Total = 56 marks		
Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.		
File this question paper in your file, ready for revision and file inspection.		